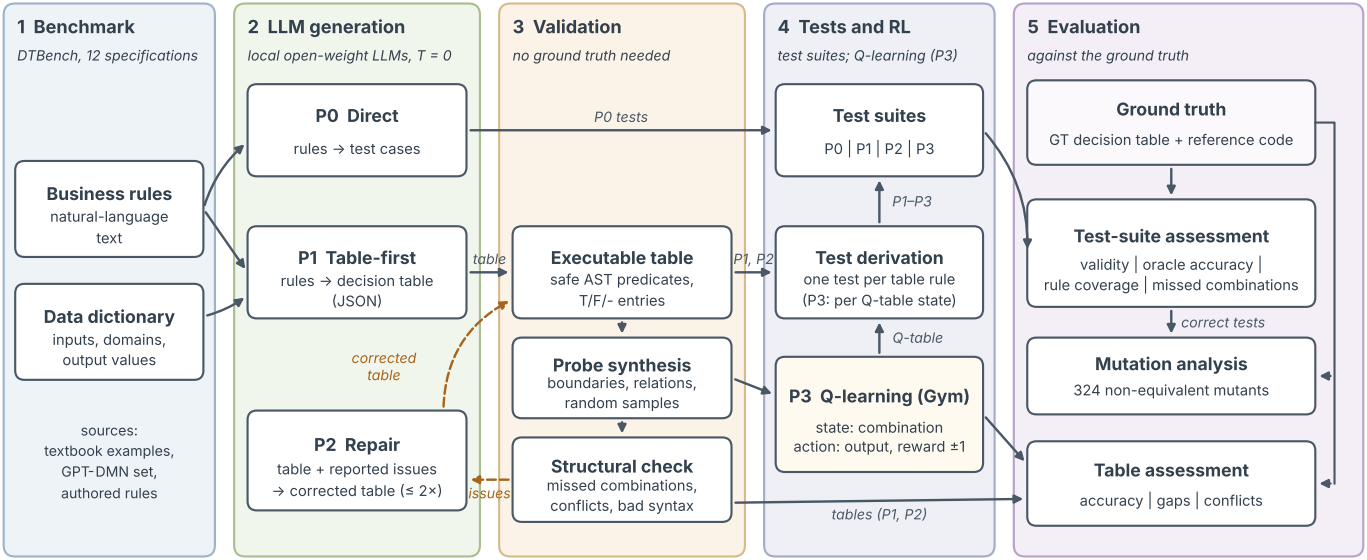
1 Benchmark
DTBench, 12 specifications
Business rules
natural-language text
Data dictionary
inputs, domains, output values
sources: textbook examples, GPT-DMN set, authored rules
2 LLM generation
local open-weight LLMs, T = 0
P0 Direct
rules → test cases
P1 Table-first
rules → decision table (JSON)
table
corrected table
P2 Repair
table + reported issues → corrected table (≤ 2×)
issues
3 Validation
no ground truth needed
P0 tests
Executable table
safe AST predicates, T/F/- entries
P1, P2
Probe synthesis
boundaries, relations, random samples
Structural check
missed combinations, conflicts, bad syntax
4 Tests and RL
test suites; Q-learning (P3)
Test suites
P0 | P1 | P2 | P3
P1–P3
Test derivation
one test per table rule (P3: per Q-table state)
Q-table
P3 Q-learning (Gym)
state: combination
action: output, reward ±1
tables (P1, P2)
5 Evaluation
against the ground truth
Ground truth
GT decision table + reference code
Test-suite assessment
validity | oracle accuracy | rule coverage | missed combinations
correct tests
Mutation analysis
324 non-equivalent mutants
Table assessment
accuracy | gaps | conflicts